

Dear Admissions Committee,

I am writing to recommend Mohamed Amine Belachhab, whom I taught mathematics in 1ère Bac Sciences Mathématiques last year and continue to teach this year in 2ème Bac Sciences Mathématiques B at Groupe Scolaire Al Jabr in Fès, Morocco. Last year, I was responsible for three mathematics sections, representing approximately 100 students. Amine was among the top five in my assessment. This year, I teach him again in my only 2ème Bac class, which has 35 students. I have therefore had the opportunity to observe him beyond a single academic year, and I am pleased that I can continue working with him during his final year of secondary school.

In 1ère Bac, Amine maintained a mathematics average above 19.8/20, earning full marks of 20/20 on numerous assessments. His performance was consistently excellent throughout the year, placing him among the top five students I taught across my three sections.

Amine has an excellent command of the mathematics we study, but some of my most memorable exchanges with him have had little to do with examinations. I remember a question he raised during a lesson on Newton's binomial theorem. I drew the class's attention to it and remarked that it reflected his interest in mathematical research. He took the question home and began investigating it independently. When he returned, he showed me work in which he had attempted to derive Taylor expansions from the binomial theorem, exploring logarithmic and arctangent functions and arriving at expressions involving π and e . I told him that it was excellent work and that he was dealing with mathematics normally taught at university.

I remember this episode because the investigation was entirely his initiative. I had not assigned him an additional project or suggested that he explore these functions. He encountered something in the lesson that interested him and decided to see how much further he could take it. When he brought the work back, our discussion had moved well beyond the original classroom question. I believe that access to more advanced mathematics, and to people with whom he can discuss it, will be particularly valuable for him.

There is another aspect of Amine's classroom behavior that I appreciate. He asks many questions, sometimes before he has completely worked out the idea he wants to express. On several occasions, he has recognized a mistake while explaining his question and corrected himself immediately. I do not regard this as a weakness. He is willing to expose his reasoning to examination, including his own, and to change it when necessary. This is a useful quality in a student who is interested in mathematics beyond the level of routine exercises. It also means that our exchanges are rarely limited to checking whether an answer is correct. He wants to understand what makes the reasoning valid and whether the same approach can be used elsewhere.

I have also followed parts of his independent investigation of Goldbach representations. He first brought me handwritten work, then a printed version, and later a PDF containing further corrections. His project explores constructive shifting methods and the propagation of representations between even integers. I want to be precise about my assessment here: I have seen his work and its successive revisions, but I have not independently verified every technical argument, nor am I presenting it as a proof of Goldbach's conjecture. What has impressed me is his continued engagement with the

problem. He has repeatedly returned to his arguments, identified difficulties, and brought back revised versions. This work is well outside our secondary-school curriculum, and he has pursued it while maintaining his academic performance.

As a mathematics teacher, I find it encouraging to see a student attempt problems of this kind and remain willing to correct his work. Amine does not yet have the mathematical training of a university researcher, and he would benefit greatly from more advanced courses and guidance. Nevertheless, he has already developed the habit of working independently on questions that interest him. I have seen this both in the relatively short investigation that followed our lesson on the binomial theorem and in the much longer process of revising his Goldbach work. His ability to recognize the limits of an argument is as important to his development as his willingness to attempt it in the first place.

I would be genuinely pleased to see Amine continue his mathematical education at your university. He has the academic preparation to engage with demanding material, and I believe that a rigorous environment, where he can encounter new ideas and have his reasoning challenged, would allow him to develop substantially. I hope you will give his application serious consideration. I recommend him with great confidence, based on the ability I have observed in my classroom and the independent mathematical work he has continued to share with me.

Sincerely,
Khalid Khrrbat
Mathematics Teacher
Groupe Scolaire Al Jabr
Fès, Morocco